WildGuard

Cross-Dataset Label Transfer for Wildlife-Trade Risk Prioritisation — Final Project Report

Group 37 · NUS-ISS Graduate Certificate in Pattern Recognition Systems · Practice Module

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Executive summary

Wildlife trade is recorded at a scale no enforcement agency can inspect record by record: the CITES Trade Database we used holds 29.2 million records since 1975. Recent complete years each contain roughly 800,000 animal-trade records (668,000–848,000 a year over 2018–2023, our count from CITES Trade Database v2026.1).

The signal that separates routine trade from trade worth a closer look rarely sits in a single field. It sits in combinations of species, product form, quantity, purpose, source and route, judged against what is normal for that species.

WildGuard is a pattern-recognition system that ranks wildlife-trade records for human review. Its central idea is **cross-dataset label transfer**. The US Fish & Wildlife Service's LEMIS data records what inspectors did with 5.5 million import line items between 2000 and 2014 (recorded in two fields: **action**: cleared = '**C'** or refused = '**R'**; and **disposition** : cleared = **'C'**, seized = **'S'**, abandoned = **'A'** or re-exported = **'R'**)

WildGuard learns the patterns that preceded enforcement action there, and applies them to the much larger CITES database, which has global coverage but no enforcement labels. A WildGuard flag on the CITES data means *this permit's records resemble trade that US officers stopped*. It is a prompt for limited human attention, not evidence that a law was broken.

To make this possible we built a knowledge model that maps both datasets onto one shared schema.

The score comes from two **supervised** tree ensembles, XGBoost and a random forest, chosen from six candidates under one protocol and combined by averaging their ranks. Five **unsupervised** detectors were tested; none improved the ranking, so the most transparent of them, a peer-deviation score, is shown to the reviewer as context instead. CITES records are grouped into export permits and ranked within each year. Every flag comes with plain-language reasons derived from TreeSHAP contributions, and a Streamlit web application serves as the review tool.

Results on held-out data (LEMIS 2013–2014, used once, after every design choice; CITES-listed species, counted per shipment):

- With an inspection budget of **5% of shipments**, **WildGuard finds 63% of the shipments with an enforcement case** (random inspection: 5%; a "most endangered first" rule: 6%). 99% of its top-1% shipments had a case, against a 3.8% base rate.
- Finding 80% of the cases requires inspecting **21% of shipments instead of 80%**, a 74% reduction in inspection workload.
- PR-AUC per line item is **0.540** (95% interval 0.519–0.559) against 0.026 for the listing rule and 0.020 for random inspection. XGBoost alone scored 0.006 higher than the chosen two-model ensemble; the

ensemble had been chosen beforehand and was kept. Restricting the model to fields CITES also records costs 0.035 PR-AUC.

- No unsupervised detector improved the ranking, and performance falls as the gap between training and scored years grows (PR-AUC 0.570 at one year, 0.512 at seven), so the deployed model is refit on all of 2000–2014.

Results on CITES (no labels). WildGuard ranks 5,465,862 CITES animal-trade records (2015–2024) in 3,274,789 export permits, after setting aside 33,053 records already marked as confiscated or for law enforcement; 32,741 permits fall in the priority tier. Three checks were fixed before the results were seen. One passes: a model that never sees the source field ranks CITES's confiscated-specimen records far above the rest (ROC-AUC 0.888). Two fail: the ranking does not clearly follow UNODC seizure statistics (Spearman ρ = 0.13 across 14 animal groups), and 30% of the 50 top-ranked taxa show no recognised risk pattern. Half of the priority tier is Siamese crocodile leather from Thailand, which US enforcement refused (95% of LEMIS imports) but which is lawful CITES trade between other countries: the clearest case of a US legal regime carried abroad.

1. Introduction

1.1 The problem

International trade in wild animals and plants is regulated under the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES). Parties report their permitted trade annually, and those reports are compiled by UNEP-WCMC into the CITES Trade Database. The full download we used (version 2026.1) contains **29,170,051 records** from 1975 to 2026. Illegal trade is common enough that UNODC's *World Wildlife Crime Report 2024* analysed about 140,000 seizures across 162 countries over 2015–2021, involving around 4,000 species (UNODC, 2024).

Enforcement agencies cannot inspect everything. The CITES Secretariat and the World Customs Organization developed *Guidance on Risk Assessment and Inspections for CITES Trade Controls* (CITES Notification No. 2025/015). CoP20 decisions in December 2025 (Decisions 20.130–20.131) encourage Parties to use and share experience with it. The guidance exists because inspection resources must be targeted, but in most agencies that targeting is still a manual, officer-driven process.

1.2 Why pattern recognition

Three properties make this a pattern-recognition problem rather than a reporting problem:

1. **Scale.** Hundreds of thousands of new records arrive each year, far beyond manual review.
2. **The signal is a pattern, not a field.**

   In February 2000 a shipment of two live, wild-caught Indian star tortoises (*Geochelone elegans*) arrived from Malaysia, declared as a personal import worth $800 (LEMIS control number 2000709917). On its own it looks unremarkable. Against its peers it does not: of the 1,063 wild-caught live *Geochelone* line items imported with a declared value in 2000–2014, 99% were commercial, typically 3–25 animals (median 10) at a median of $35 per animal. This one was personal-purpose and priced at $400 per animal, 11 times the median. Personal-purpose imports of live *Geochelone* were refused 32% of the time, against 3.5% for commercial ones. This shipment was seized. Detecting such a case requires comparing each record with its peers across several fields at once.[1]

[1] Our first presentation quoted peer figures (26 peer shipments of 19–250 animals at a median of $6) from an exploratory LEMIS sample. The figures here are recomputed on the full dataset.

3. **Labels exist, but in the wrong place.** The outcomes needed to learn what risk looks like exist in one national dataset (LEMIS) but not in the global one (CITES).

1.3 Objective and research questions

Objective: use real enforcement outcomes recorded in one national wildlife-trade dataset (LEMIS) to build a statistically grounded system that ranks records for human review in the much larger, unlabelled global CITES database.

We framed the project around six questions, so that its value is measured rather than asserted:

#	Question	Where it is answered
RQ1	Can LEMIS and CITES be aligned so that the same trade looks the same in both, and what cannot be aligned?	§3.4
RQ2	Does it work — can enforcement outcomes be predicted on shipments the model never saw, overall and on the CITES-like part of LEMIS (CITES-listed species, commercial trade)?	§6.2
RQ3	Is it useful — how much inspection effort does it save compared with random or rule-based inspection, measured on held-out LEMIS (CITES has no outcomes)?	§6.3
RQ4	Does combining supervised and unsupervised evidence help?	§6.4
RQ5	Does it transfer — which parts of the shift from LEMIS to CITES can be measured, how much does each cost, and which adaptation steps help?	§6.6
RQ6	What does it surface in CITES, and is that plausible against independent evidence?	§6.7

1.4 What a flag means

WildGuard is a **triage aid**. The label it learns is an enforcement outcome. That outcome is often a paperwork refusal, not proof of trafficking, and LEMIS reflects US enforcement priorities. On CITES no ground truth exists, so the system *ranks* records within an inspection budget instead of claiming calibrated probabilities. Every output is meant to be reviewed by a person. An *inspection budget* is the share of records or permits a team can review in a period: a budget of 5% means the top 5% of the ranking is reviewed and the rest is not. The priority tier is a budget of 1% of each year's permits, the review tier the next 4%.

A flag also means less than it may seem. Most US enforcement cases never appear in CITES at all: when LEMIS outcomes are matched to the CITES records the US reported for the same 2013–2014 imports, 72% of cleared line items have a match but only 39% of enforcement cases, because a seized or abandoned shipment never entered the country (notebook 07). A flag therefore says that a CITES record resembles trade that was stopped in the US, not that the record itself was stopped.

2. Project journey: from image fusion to label transfer

Our submitted proposal (15 September 2026) described a hybrid system that fused structured CITES and seizure records with visual recognition of pangolin products. While preparing the first presentation we re-examined that design and dropped it, for three groups of reasons (Figure 1):

- **Modality and fairness.** The visual stream covered one species (pangolins); the structured stream covered all CITES taxa. Treating missing images as weaker evidence would systematically suppress scores for every non-pangolin case, and positive visual evidence would systematically inflate pangolin cases, independent of true risk. The two streams were not even guaranteed to describe the same shipment. Proving that "hybrid beats single-stream" would have been hard for one species and impossible across all of them. Published work on modality imbalance and "greedy" multimodal learning shows naive fusion can lose to the best single modality when one modality is sparse (Wang et al., 2020; Wu et al., 2022; Peng et al., 2022).
- **Usefulness.** The system would mostly have restated what is already known ("pangolins are heavily trafficked") rather than saying which *specific* record deserves attention now.
- **Data feasibility.** The candidate pangolin image datasets were available only on request; our request to Cardoso et al. received no response. Perhaps this could have been addressed with conditional GANs but we ultimately decided to drop this project direction as synthetic augmentation of a small seed set would have introduced a synthetic-to-real domain gap.

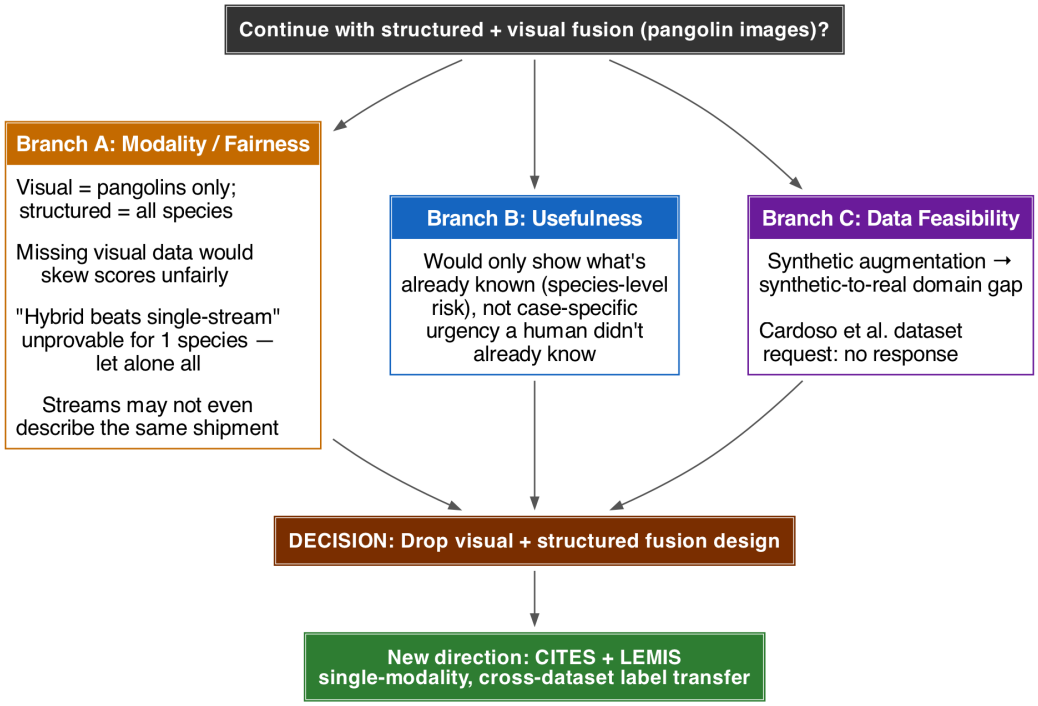


Figure 1 – Decision path from the original proposal to the current direction (presented at the first presentation, 30 September 2026).

The new direction keeps a single modality (structured trade records) and uses two large, real, freely downloadable datasets that complement each other's gaps: **LEMIS has labels but only US scope; CITES has global scope but no labels.** In our literature search we found no published work that transfers LEMIS enforcement outcomes to CITES records. The closest precedent was an informal analysis that applied LEMIS labels to commercial shipping records rather than to CITES. The contribution is therefore an *application* of established techniques in a new combination, not a new algorithm.

3. Data

3.1 Sources

	LEMIS (US Fish & Wildlife Service)	CITES Trade Database
Content	Every wildlife import declared to USFWS, whether or not the species is CITES-listed	Trade in CITES-listed species, reported by all Parties
Version	Zenodo record 3565869, v1.1.0 (Eskew et al., 2019; 2020); checksum verified	Full download v2026.1, extracted 9 June 2026 (UNEP-WCMC)
One record is	One product line item of a US import declaration	One line item of a permit, as reported by one Party (exporter or importer); 97% carry an export permit ID
Size	5,512,706 line items in 2,088,676 shipments	29,170,051 records (1975–2026), 20,962,017 of them animals
Period used	2000–2014; the model learns from animals with a class (5,107,119 line items)	Animal records 2015–2024 (8,247,093), of which 5,465,862 are scored (§3.4); 2013–2014 US-reported records for the alignment check
Labels	`action` (C cleared, R refused); `disposition` (C cleared, S seized, A abandoned, R re-exported)	none
Declared value	yes (USD), but missing for 2008–2010 and 2014	no

Both datasets were checked for updates during the project. LEMIS v1.1.0 is the latest Zenodo version and a one-off release. CITES v2026.1 is the current annual release. It contains the annual reports received by 5 May 2026, so its most recent years are incomplete. WildGuard therefore ranks 2015–2024 and leaves 2025–2026 out (2025 holds about 10,000 animal records against roughly 700,000 in a complete year); the app's preview of the CITES data shows records from the same window.

Before designing the alignment, we explored both datasets as published, following their publishers' documentation (Eskew et al., 2020; CITES Secretariat & UNEP-WCMC, 2022). The exploration is in three notebooks (`SYSTEM CODE/notebooks/`, 01–03). It took as given only the publishers' rules for reading each dataset, recorded in `DATA/USING_LEMIS_AND_CITES.md`. The preprocessing and alignment rules, in `DATA/ALIGNING_LEMIS_AND_CITES.md`, were then written from its findings. Two design decisions the exploration left open were then settled by experiments with decision rules written before the results were seen (notebooks 04 and 05). Sections 3.2 and 3.3 report what the exploration found; §3.4 describes the alignment and the two decisions.

3.2 LEMIS labels and class imbalance

LEMIS records what inspectors did with each line item in two fields. `action` records whether the item was cleared (C) or refused (R). `disposition` records what then happened to it: cleared (C), seized (S), abandoned (A) or re-exported (R). The letter R therefore means *refused* in one field and *re-exported* in the other.

`action`	`disposition`	Line items	Label
C	C	5,417,030	cleared
R	S	53,028	enforcement case
R	A	22,257	enforcement case

action	disposition	Line items	Label
R	R	18,212	enforcement case
R	C	2,075	enforcement case
other combinations		104	by the rule below

We label a line item an enforcement case if it was refused (`action` = R) or its disposition was seized, abandoned or re-exported (`disposition` = S, A or R). This gives 95,665 positives (1.74%): 53,030 seized (55%), 22,257 abandoned (23%), 18,235 re-exported (19%) and 2,143 refused with another or missing disposition (2%). These are what US inspectors did, not findings that a law was broken. Refusals include paperwork problems, and illegal shipments that went undetected appear as cleared.

The positive rate is stable across years (1.5–2.2%), which makes a temporal split meaningful. Labels are effectively shipment-level: only 1.0% of multi-item shipments have mixed outcomes. A shipment's exporter, date, port and companies are also identical on all its line items, so a shipment's line items are always kept in the same split.

At a 1.7% positive rate accuracy is uninformative, because clearing everything scores 98.3%. We therefore evaluate with ranking and budget metrics (§6.1).

3.3 What the raw data shows

LEMIS

Group	Enforcement rate	Group	Enforcement rate
Purpose: personal (P)	25.5%	Source: confiscated (I)	74.6%
Purpose: commercial (T)	0.98%	Source: unknown (U)	25.6%
Purpose: hunting trophy (H)	0.55%	Source: wild (W)	1.76%
Product: caviar	29.5%	Source: captive-bred (C)	0.56%
Product: extracts / medicine	28.0%	Appendix I in force	4.0%
Product: ivory	12.0%	Appendix II in force	1.8%
Product: live animals	0.52%	No CITES listing found for the species	1.5%

Appendix in force: the most recent appendix CITES recorded for the same species in or before the shipment year (Appendix N excluded).

Half of all enforcement cases (49.6%) are personal imports, typically wildlife products in travellers' luggage: wild-sourced personal "extracts/medicine" is refused 96% of the time. Commercial imports account for 41.9% of cases.

The exploration also found properties of the data that preprocessing has to respect:

- **Missing values.** These are written as NA. Codes that EcoHealth Alliance could not interpret are written as `non-standard value` (mainly in purpose, 12,794 rows, and source, 667). Both are treated as missing.
- **Duplicates.** 153,485 rows (2.8%) are exact duplicates. EcoHealth Alliance kept them on purpose because they can be real, for example 25 identical garments declared on separate lines. We keep them too.
- **Declared value.** Value is missing for all of 2008–2010 and 2014, and for 30% and 44% of 2011 and 2013. Where it is recorded, a zero value is more than twice as common among enforcement cases (24.1%) as among cleared items (9.8%). Value is therefore a LEMIS-only signal, and it is absent from the 2014 test year.
- **Quantities.** Quantities are heavy-tailed within each of the 12 units (counts: median 10, 99th percentile about 36,000), and the unit MT means metres, not tonnes.

- **Taxonomy.** 7.3% of rows have no class. genus sometimes holds a higher taxon (the coral order *Scleractinia*, 45,922 rows; the families *Cheloniidae* and *Elephantidae*) or a placeholder such as "Non-CITES entry".

CITES

- **Double reporting.** Each record is reported by one Party, and the same trade is often reported by both the exporter and the importer. For animals in 2015–2024 there are 4,836,000 exporter-reported and 3,411,093 importer-reported records. Counts and shares double-count trade unless one reporter type is used. Below we use exporter-reported records, which are the more complete.
- **Permits link the two reports.** The export permit ID, a random code that replaces the real permit number, is present on 97% of records and is the same on both sides of a trade. 1.47 million permits are reported by both countries, and the two reports agree on exporter, importer and species in over 97% of cases. Matching on field values alone pairs almost nothing one-to-one (0.8%), because one combination of year, species and countries usually covers many permits. The IDs are not unique to one trade, however: the same ID also occurs on unrelated trades between other countries, probably where placeholder permit numbers were anonymised to one ID, so WildGuard identifies a permit by its ID, exporter, importer and year (§4.7).
- **One-sided reporting.** 6% of importer-reported records have no exporter-reported record of the same trade; for 58% of them the exporting country filed no export records that year (for example Tonga and the Cook Islands). Conversely, 23% of exporter-reported records have no importer-reported counterpart. This gap is concentrated in a few importing countries: Taiwan (not a CITES Party), Russia and Canada each leave over 95% unconfirmed. Whether a trade is reported by both sides therefore says more about the countries' reporting than about the trade.
- **Reporting lag.** Recent years are incomplete because annual reports arrive late. Complete years hold 668,000–848,000 animal records (2018–2023), but 2024 has 686,852 and 2025 only 10,040. Records are ranked within each year, so the gap does not distort the ranking.
- **Appendix N.** 6,511 records in Appendix N are non-CITES species reported by EU members under stricter domestic measures. They are not CITES listings and are excluded.
- **Trade mix.** Trade is overwhelmingly commercial: purpose T in 90.4% of records, personal in 3.1%. Reptile leather goods dominate: "leather products (small)" are 56.4% of records and reptiles 65%.
- **Source I.** Source I ("confiscated or seized specimen") does not record a seizure. It marks the legal movement of an item seized earlier, such as its return to the country of origin. Seizures are held in the separate, access-restricted CITES Illegal Trade Database.
- **Units.** A blank unit and "Number of specimens" both mean a count. The same term is often recorded in several units: "specimens" appears in 13, and a third of "bodies" are recorded in kilograms.
- **Higher taxa.** 86,568 records leave Genus blank. 81,773 of them are reported above genus level and write the higher taxon in Taxon, for example "Scleractinia spp." (60,899) or "Elephantidae spp." (9,818); 4,599 are hybrids.
- **Origin.** A blank origin (31% of records) means the exporter is the origin.

What changes between the two datasets

	LEMIS 2000–2014	CITES 2015–2024 (exporter-reported)
Purpose: commercial	74.5%	90.4%
Purpose: hunting trophy	17.6%	3.0%
Purpose: personal	3.4%	3.1%
Source: wild	78.0%	60.5%
Source: bred or born in captivity (C, F)	16.7%	31.4%
Class: reptiles	13.1%	65.1%

	LEMIS 2000–2014	CITES 2015–2024 (exporter-reported)
Class: mammals	28.8%	8.2%
Product: leather products (small)	10.5%	56.4%
Product: shells	11.1%	0.03%
Species name found in CITES	28.7%	100%

The model is trained on one population and applied to a very different one. Personal imports supply half of LEMIS's enforcement cases but only 3% of CITES records. We therefore also report a commercial-only evaluation (§6.5). Most features describe a record relative to its peers rather than in absolute terms (§4.2), and we measure the shift directly in §6.6.

Where the two datasets should agree, they do. The US compiles its CITES annual report from the same declarations that LEMIS holds. For 2013–2014, LEMIS line items of species found in CITES (243,172) and CITES records reported by the US as importer (246,206) are almost equal in number. Their purpose mix is also close: personal 3.29% vs 3.27%, scientific 0.92% vs 0.91%, commercial 78% vs 84%. They differ mainly in taxonomic class (corals 29% vs 45%), which reflects how the two systems record taxa and products rather than different trade.

3.4 Schema alignment — the knowledge model

LEMIS and CITES describe the same kind of event in different vocabularies, and §3.3 showed that some codes that look the same are not. We map both onto one shared schema with an explicit crosswalk (`DATA/ALIGNING_LEMIS_AND_CITES.md`, §2.1):

Field	LEMIS	CITES	Shared form
Records used	animals with a class (decision below); NA and `non-standard value` = missing; placeholder genera = unknown	animal records (class present), Appendix N removed; source I and purpose L records not ranked; mirror reports handled by rule D (decision below)	—
Product form	94 three-letter codes (92 in use)	101 English terms for animals (81 in 2015–2024)	shared product groups, defined by how each vocabulary records an object (live vs dead, raw vs worked; ivory, horn and bone kept apart); one leather group for small and large goods, shoes and trim; live rock with raw coral
Unit	NO, KG, GM, MG, LT, ML, C3, M2, C2, M3, MT (metres), CM	words; blank = count; also microgrammes	count / kg / litre / m / m^2 / m^3, quantities converted
Source	W, C, F, R, D, I, U, A; pre-Convention = **P**	W, C, F, R, D, I, U, X; pre-Convention = **O**	CITES codes; LEMIS P → O; CITES X (no LEMIS equivalent) and missing → unknown
Purpose	T, H, P, S, E, Z, M, Q, B, G, L; reintroduction = **Y**	same letters; reintroduction = **N**	CITES codes; LEMIS Y → N; missing → unknown

Field	LEMIS	CITES	Shared form
Country	ISO-3166 alpha-2, plus VS, ZZ, ZR, TP, AN, XX	ISO-3166 alpha-2, plus XV, HS, XX; blank origin = exporter	ISO codes (VS → XV, ZZ → HS, ZR → CD, TP → TL), plus world region
Class	Catalogue of Life names (Actinopterygii, Sarcopterygii)	CITES names (Actinopteri; Dipneusti and Coelacanthi)	CITES names
Higher taxa	order or family name in `genus` (64,805 rows)	`Genus` blank, `Taxon` = "*name* spp." (81,773 records)	genus field holds the higher taxon in both
CITES appendix	not recorded	recorded	for LEMIS taxa, the listing in force in the shipment year: the most recent appendix CITES recorded for the species (its genus only when no species is named; a named species CITES never recorded is marked "genus listed") in or before that year; Appendix N ignored
Permit	none	export, import and origin permit IDs	not a model feature (LEMIS has none); the export permit ID is kept to group and compare records
Declared value	USD	—	used only by the LEMIS-only "full" model

Three parts of the crosswalk **correct codes** that look identical but are not:

- **Pre-Convention.** LEMIS records pre-Convention specimens as source P, while CITES uses O and has no source P. Without the mapping, every CITES pre-Convention record would carry a code the model never saw in training.
- **Reintroduction.** Reintroduction into the wild is purpose Y in LEMIS and N in CITES.
- **Product form.** The two product vocabularies divide objects differently. LEMIS has separate codes for shoes, garments and trim, while CITES records most such items as "leather products (small)", which make up 56% of CITES records but only 10% of LEMIS. Product groups are therefore defined by how each vocabulary records an object, not by matching names. LEMIS shoes therefore join one leather group with small and large leather goods, and live rock joins raw coral (notebook 06).

Two decisions were left open by the exploration. Each was settled by an experiment whose decision rule was written before its results were seen.

Which CITES records are scored (notebook 05). Aggregate statistics on CITES (§6.7) use exporter-reported records only, as the CITES guide advises. For the review queue, four options were compared by the share of the model's top 1% they waste:

- **exporter-reported only** misses 14.7% of that top 1%: high-risk trade that only the importer reported;
- **both reporter types** spend 18.7% on second reports of trades already in the queue;

- **merging each trade's two reports into one record** is not possible reliably: even with permit IDs, only 33.7% of exporter records pair one-to-one;
- **rule D** loses 4.1%.

Rule D scores every exporter-reported record, plus importer-reported records with no exact exporter-reported counterpart (same year, species, term, unit, countries, origin, purpose and source). This leaves 5,465,862 records to score. The ranking of the options is the same whichever training population is used. Permit IDs are not used to drop importer reports: when both countries report a permit, the model often scores the two reports differently, and dropping one loses the flagged report more often than it removes a duplicate. Instead, the export permit ID is kept so that the review tool can group a consignment's records and show both countries' reports side by side. These figures come from a rerun on the rebuilt data with the final model settings, in which rule D had the lowest cost in all four cases tested.

Which LEMIS records the model learns from (notebook 04). Four training populations were compared on the same held-out records, LEMIS line items of species with a CITES listing in force, which are the records most like CITES:

- all line items;
- animals with a class;
- CITES-listed species only;
- listed species and genera.

The decision was first taken on 2013–2014 with the pipeline's earlier settings, before the year roles of §4.3 existed. It was then checked on the rebuilt data and the final model settings, evaluated on the selection year 2011 so that the test years stay used once. Recall at a 1% review budget was indistinguishable (0.383–0.386; all paired-bootstrap intervals include zero), so the rule as written keeps **animals with a class**, which drops plants and records without a taxonomic class (placeholders such as "Non-CITES entry"). Judged on PR-AUC, the design's deciding metric (§4.3), training on CITES-listed species only was slightly better: +0.007 on listed line items (interval +0.002 to +0.012) and +0.017 on listed commercial ones. The check came after the one test of the finished system, which used animals with a class, so the design was not changed; training on listed species only is future work (§9). The choice matters in deployment: the CITES top 1% under the two populations overlaps by 79%.

After mapping, fewer than 0.05% of records in either dataset fall outside the shared product groups (mostly plant products in LEMIS). A further 0.5% of LEMIS and 0.02% of CITES records are recorded only as "unspecified". Three checks show whether the alignment holds. First, no CITES value of a shared field should be missing from the LEMIS training data; this is the test that exposed the pre-Convention, reintroduction, unit and country mismatches. Second, product-group shares are compared in both datasets, since a group that is large on one side and empty on the other points to a coding difference rather than different trade. Third, the domain classifier and population stability index of §6.6 B measure how far the aligned datasets still differ: no training population is close to CITES (domain-classifier AUC 0.96–0.99), so restricting the training data is not a substitute for the relative features of §4.2.

4. System design

4.1 Architecture

WildGuard works in two lanes (Figure 2). In the LEMIS lane, where enforcement outcomes exist, records are selected, aligned, described relative to their peers, and used to train and test the models. In the CITES lane the same crosswalk is applied, records are described with the LEMIS peer norms, scored by the models refit on all LEMIS years, grouped into export permits and ranked within each year for review. Labels exist only in the first lane, so every model choice and the one test are made there; the CITES lane is checked against outside evidence instead (§6.7).

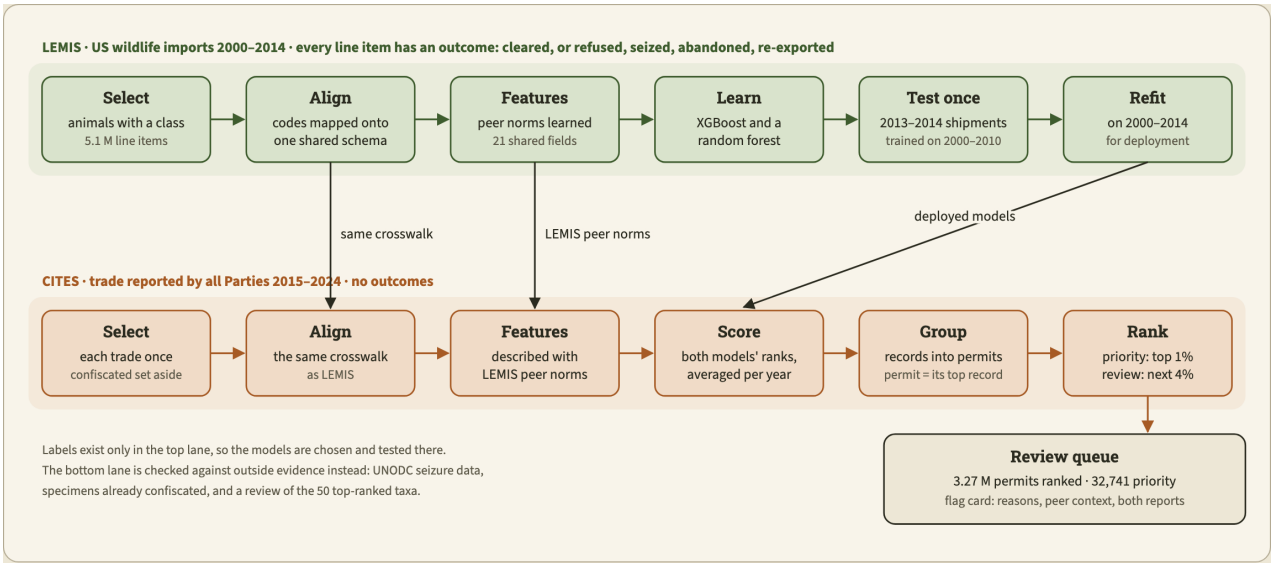


Figure 2 – WildGuard architecture: what crosses from LEMIS, where outcomes exist, to CITES, where they do not.

The pipeline is implemented as a Python package (`SYSTEM CODE/wildguard/`) and runs end to end with one script (`run_pipeline.sh`). DuckDB streams the ~6 GB of raw CSV into aligned Parquet files; all modelling uses pandas, scikit-learn and XGBoost. Every design choice, with its evidence, is recorded in `SYSTEM CODE/DESIGN_QUESTIONS.md` and the design notebooks 06–14.

4.2 Relative risk features

The worked tortoise example shows that risk lies in *deviation from a record's peers*. Most features are therefore relative, and relative features also transfer better across datasets than absolute ones. "Ten times the usual quantity for this species and product" means the same in LEMIS and CITES even though their absolute volumes differ.

Feature	Definition
`qty_z`	Robust z-score of log quantity within the peer group (genus × product group × unit), using median and MAD; backs off to class level for groups under 20 records
`unit_value_z` (LEMIS only)	Robust z-score of log declared value per unit within the peer group
`route_share`, `route_n_log`	Share of the genus's records exported by this country; route volume per million records
`combo_rarity`	$-\log_{10}$ frequency of the product × purpose × source combination
`purpose_share_in_genus`, `source_share_in_genus`	How common this purpose / source is for this genus
`peer_n_log`	Size of the peer group, per million records (thin groups are less certain)
`reexport_flag`, `is_wild`	Origin ≠ exporter; source = wild
Categorical	class, product group, unit group, purpose, source, CITES appendix, origin/exporter region, top-300 genus, top-120 origin and exporter countries

Peer norms are learned from a reference population, the LEMIS training years, and CITES records are described against the same LEMIS norms (those of the deployed model's years 2000–2014). Learning the norms from CITES itself, from rule D's records, was the earlier design. Tested on held-out LEMIS populations, norms learned from the target were significantly worse on two of six (Oceania −0.066 PR-AUC, the time holdout −0.007), so it was dropped (notebook 08). The genus and country vocabulary always comes from the training data, and counts are expressed per million records so that they mean the same thing in populations of different size.

Two feature sets are trained in parallel:

- **Shared**: the 21 fields both datasets record; this is the deployed model. Feature groups were selected by leaving one group out on 2011: removing any group significantly lowered PR-AUC (most: countries −0.023, quantity −0.012), and a consignment feature (the number of distinct sources per shipment or permit) added nothing once the random forest was in (notebook 12).
- **Full**: adds the LEMIS-only declared value, unit value, entry port and month. It is used only to measure what alignment gives up (§6.5) and to rank US import declarations in the app.

4.3 How the models were chosen

Models were chosen by comparison rather than by assumption. Every step was decided on LEMIS, where labels exist, with its decision rule written before its results were seen, and the test years were used once, at the end (Figures 3 and 4).

- **Year roles.** Training uses 2000–2010, with hyperparameters tuned by time-ordered cross-validation inside those years (train to 2006, validate on 2007–2008; train to 2008, validate on 2009–2010) and peer norms refitted on each fold's training part. Every design choice is made on 2011; 2012 is held back for the triage bands; 2013–2014 are the one test. A shipment's line items share a year, so they never fall on both sides of a split.
- **The deciding metric** is PR-AUC on line items of species with a CITES listing in force, the part of LEMIS most like CITES. Accuracy rewards clearing everything (98.3%), ROC-AUC flatters rare positives, and F1 needs a threshold that the CITES ranking does not have. Recall at a 1% budget was considered first, but cases are 2.4% of listed line items, so a 1% budget cannot reach more than about 42% of them and strong models are squeezed together. Differences are judged by a paired bootstrap that resamples whole shipments.
- **Candidates.** Logistic regression (trained by L-BFGS and by stochastic gradient descent, compared like for like), naive Bayes, a decision tree, a random forest, AdaBoost and XGBoost. A linear SVM differs from logistic regression only in its loss; an MLP and a kernel SVM cost too much at five million rows; and nearest-neighbour scoring of 5.5 million CITES records against five million stored ones is impractical, so these were left out. A domain-adversarial network, suggested for the shift between the datasets, was not pursued: without an MLP baseline, a gain from adversarial training could not be told apart from a gain from using a neural network.
- **Search.** Five designed configurations per candidate: four compare the class-imbalance options (none, class weights, random undersampling, SMOTE-NC, or the balanced random forest and RUSBoost), and the fifth changes the candidate's main setting using the best of them.

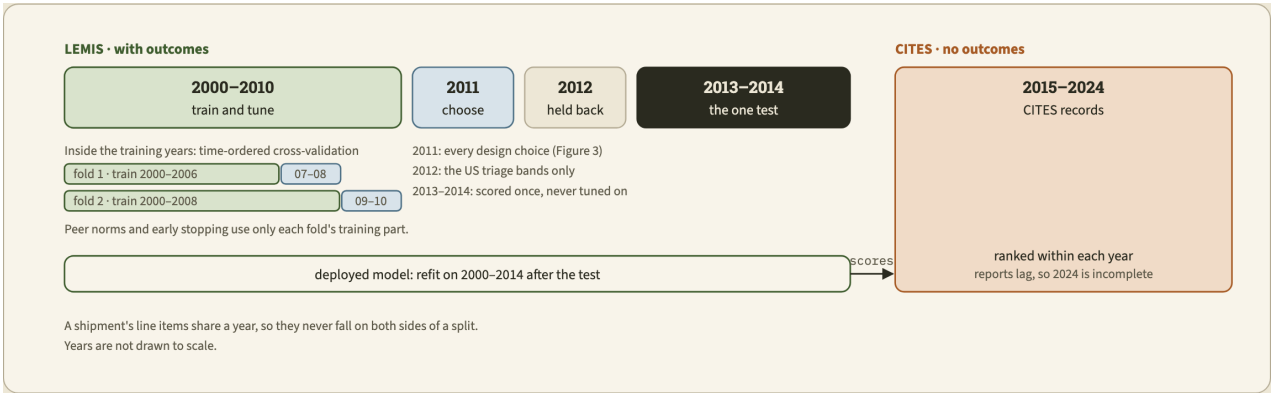


Figure 3 – Which years each step may see.

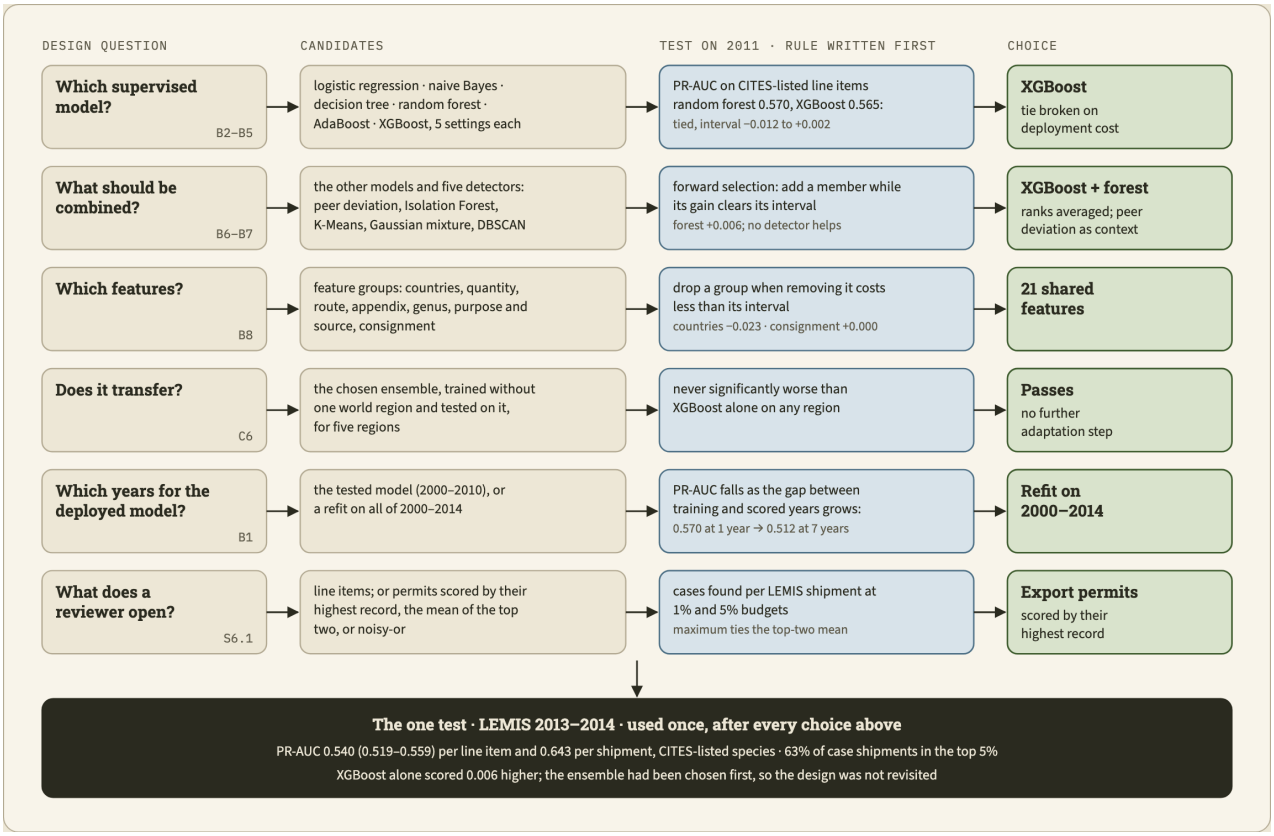


Figure 4 – How each model choice was made: the question, the candidates, the test on 2011 with its rule written first, and the choice.

What the comparison found (notebooks 10–14):

- **Supervised model.** On listed 2011 line items the random forest (PR-AUC 0.570) and XGBoost (0.565) were tied (difference −0.005, interval −0.012 to +0.002); every other candidate was significantly worse (logistic regression −0.073, decision tree −0.113, AdaBoost −0.156, naive Bayes −0.193). The written rule sends ties to the simpler model, the random forest. After seeing the tie we chose XGBoost on deployment cost instead: the forest would have added roughly 340 MB to the hosted app and needed an extra package for its reasons, while XGBoost is 1 MB with exact contributions built in. We state this because it is a choice made after a result.
- **Class imbalance.** For every strong candidate, no imbalance handling ranked best; class weights lowered PR-AUC in the cross-validation (XGBoost 0.570 → 0.559, random forest 0.556 → 0.490). Reweighting

moves a decision boundary but does not improve the order of the records, which is all that PR-AUC and the ranking use.

- **Combining.** Forward selection from XGBoost added the random forest (+0.006 at the deployed size of 100 trees of depth 16, interval +0.003 to +0.010) and then stopped: no unsupervised detector and no other model improved the ranking. A fitted logistic meta-learner tied with a simple average of the two models' percentile ranks, so the average is used.
- **Transfer and age.** Trained without one world region and tested on it, the ensemble was never significantly worse than XGBoost alone on any of five regions. Performance falls with the gap between training and scored years: five-year training windows ending 1, 3, 5 and 7 years before 2011 reach PR-AUC 0.570, 0.536, 0.528 and 0.512 (notebook 13). The deployed model is therefore the tested configuration refit on all of 2000–2014.
- **Review unit.** Combining a shipment's line scores by their maximum tied with the mean of the top two and beat noisy-or on 2011 LEMIS shipments. At a 1% budget of shipments, 410 of 411 review slots reached a shipment with a case, while a 1% budget of line items spent slots on repeat lines of the same shipments (notebook 14). The rule for combining a consignment's records can therefore be tested on LEMIS, through shipments.

4.4 Supervised learning

- **XGBoost**: gradient-boosted trees with native categorical support; depth 6, learning rate 0.05, minimum child weight 20, row and column subsampling 0.8, no class weights, 357 boosting rounds (the mean best round of the cross-validation folds).
- **Random forest**: 100 trees of depth 16, minimum leaf size 5, square-root feature sampling, no class weights (27 MB; 200 deeper trees gained only 0.003 more and would not fit the hosted app's memory).
- **Logistic regression** (one-hot categories, z-scored numerics, L-BFGS) is kept as the linear comparison model.

The score is the average of the two models' percentile ranks. On CITES the ranks are taken within each year, so a score means the same in every year although reporting volumes differ.

4.5 Unsupervised detectors

Five detectors were tested, each fitted without labels: the peer-deviation score (how far a record's quantity, route, purpose, source and product combination sit from its peers'), an Isolation Forest, K-Means distance (8 clusters, silhouette 0.29), a Gaussian mixture (12 components by BIC) and DBSCAN. On their own all rank enforcement cases above random (PR-AUC on listed 2011 line items: peer deviation 0.151, Isolation Forest 0.111, DBSCAN 0.094, Gaussian mixture 0.086, K-Means 0.038; random 0.024), but far below the supervised models (XGBoost 0.565), and none improved the ensemble once XGBoost and the random forest were in it. A great deal of legal trade is statistically unusual: unusual is not illegal. The peer-deviation score, the strongest and most transparent detector, is therefore shown on the flag card as context for the reviewer, not added to the score.

4.6 Explanations and the flag card

For every flagged record, XGBoost's exact per-feature contributions (TreeSHAP) identify the inputs that pushed its score up, and the top three become plain-language reasons an analyst can check against the record. For the tortoise shipment of §1.2 these are *"declared purpose 'Personal' – only 5.9% of Geochelone records"* and *"rare route: Malaysia accounts for 0.07% of recorded Geochelone trade"*. The reasons explain XGBoost, one of the two models, and the flag card says so. Beside them the card shows the peer-deviation context, the permit's other records with both countries' reports where they differ, and a warning where the pattern may reflect US law (Figure 5).

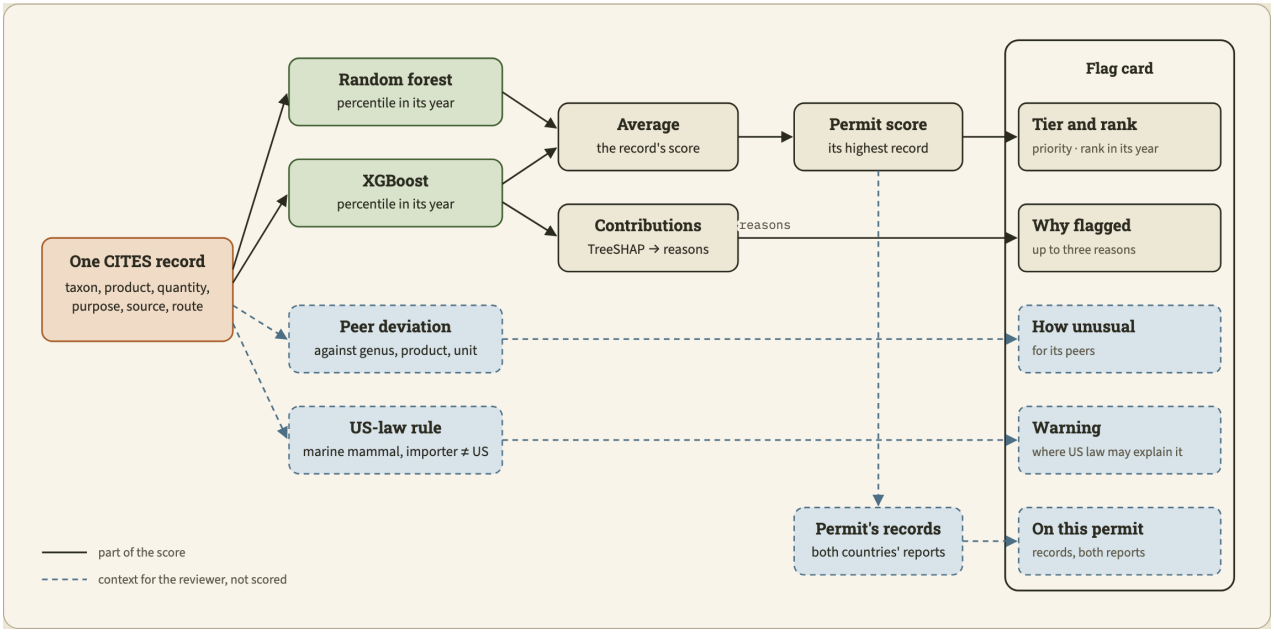


Figure 5 – What makes up a score, and what the flag card adds.

4.7 Deployment on CITES

1. **Records.** CITES animal records 2015–2024 (8,247,093) are mapped with the same crosswalk, and each trade is scored once under rule D (§3.4): 5,498,915 records. Appendix N records are removed: they are species that EU members reported under stricter domestic rules, not CITES listings, and they exist only because one region chose to report them. The same reasoning removes N from LEMIS. Because LEMIS has no appendix field, an earlier version of the pipeline copied each species' latest CITES appendix onto LEMIS, sometimes N, and 391,241 LEMIS line items (7.1%) carried an appendix that means nothing for a US shipment. The listing in force in the shipment year, with N ignored, replaced it.
2. **Features.** CITES records are described against the LEMIS peer norms of the deployed model (§4.2), with the vocabulary fixed to the training data's.
3. **Set aside.** Records already marked as enforcement material, source I (confiscated or seized specimen) or purpose L (law enforcement, judicial or forensic), are not ranked (33,053): telling an analyst to review what has already been seized adds nothing. 5,465,862 records are ranked.
4. **Scored.** Both models, refit on 2000–2014, score every record, and their percentile ranks within the year are averaged.
5. **Grouped into permits.** Records sharing an export permit form one consignment, scored by its highest record (§4.3). A permit is identified by its ID, exporter, importer and year: the anonymised IDs also occur on unrelated trades, and grouping by ID alone had merged them. A record without a permit stands alone. This gives 3,274,789 permits.
6. **Ranked.** Within each year, permits are cut into tiers by inspection budget: **priority** = top 1%, **review** = next 4%, **routine** = the rest (Figure 6). Without CITES labels the score cannot be calibrated as a probability, so budget tiers are the defensible output: 32,741 priority permits over ten years, about 3,300 a year worldwide.

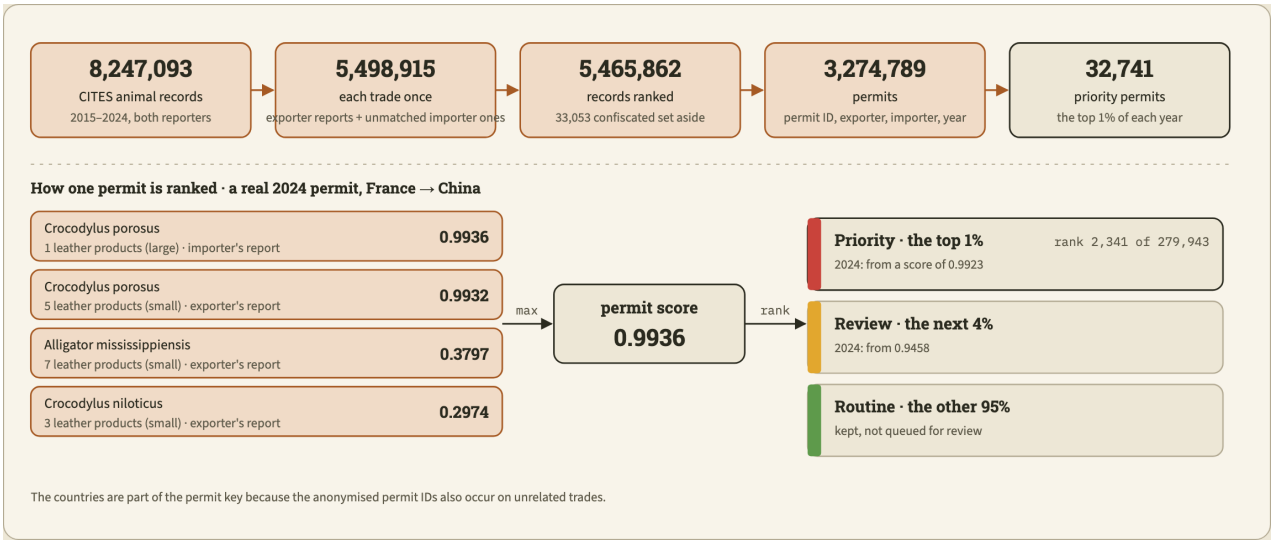


Figure 6 – From CITES records to the review queue, with a real 2024 permit.

4.8 MVP: the WildGuard review tool

The Streamlit application (`SYSTEM CODE/app/app.py`) runs from a clone of the repository without the raw data. Each page answers one question, leads with a plain answer and a comparison, and keeps the measurement detail in expanders:

Page	What it shows
Overview	What WildGuard does, its headline results, what a flag means, how it works in seven steps, and the two source datasets as published
Review queue	CITES export permits ranked within each year (priority top 1%, review next 4%). A permit's flag card shows its tier and rank in the year, the reasons, the peer context, its records with both countries' reports, and the US-law warning where it applies (Figure 7)
Check a record	Ranks one trade record against that year's CITES permits, or a US import declaration with declared value, port and month against US shipments 2000–2014 (Figure 8)
Does it work?	The one test: an inspection-budget slider over held-out LEMIS shipments, cases found against random and rule-based inspection, every model compared, what the US-only fields add, triage bands and the measured shift
What it flags	The three CITES checks with their pass marks and results, and where priority permits concentrate

Risk is shown as a tier and a rank in the year (for example "top 0.01% of 2024 permits"), never as a probability, because on CITES the score is a ranking. Hosted, the app uses about 600 MB of memory.

Access. The app is hosted on Streamlit Community Cloud at https://wildguard-group37.streamlit.app. It opens in any web browser with no account or installation; Appendix A gives step-by-step instructions.



Figure 7 – Review queue: a 2024 priority permit for sperm-whale carvings from Greenland to Denmark, with its reasons, the US-law warning and both countries' reports, which disagree (carvings against teeth).

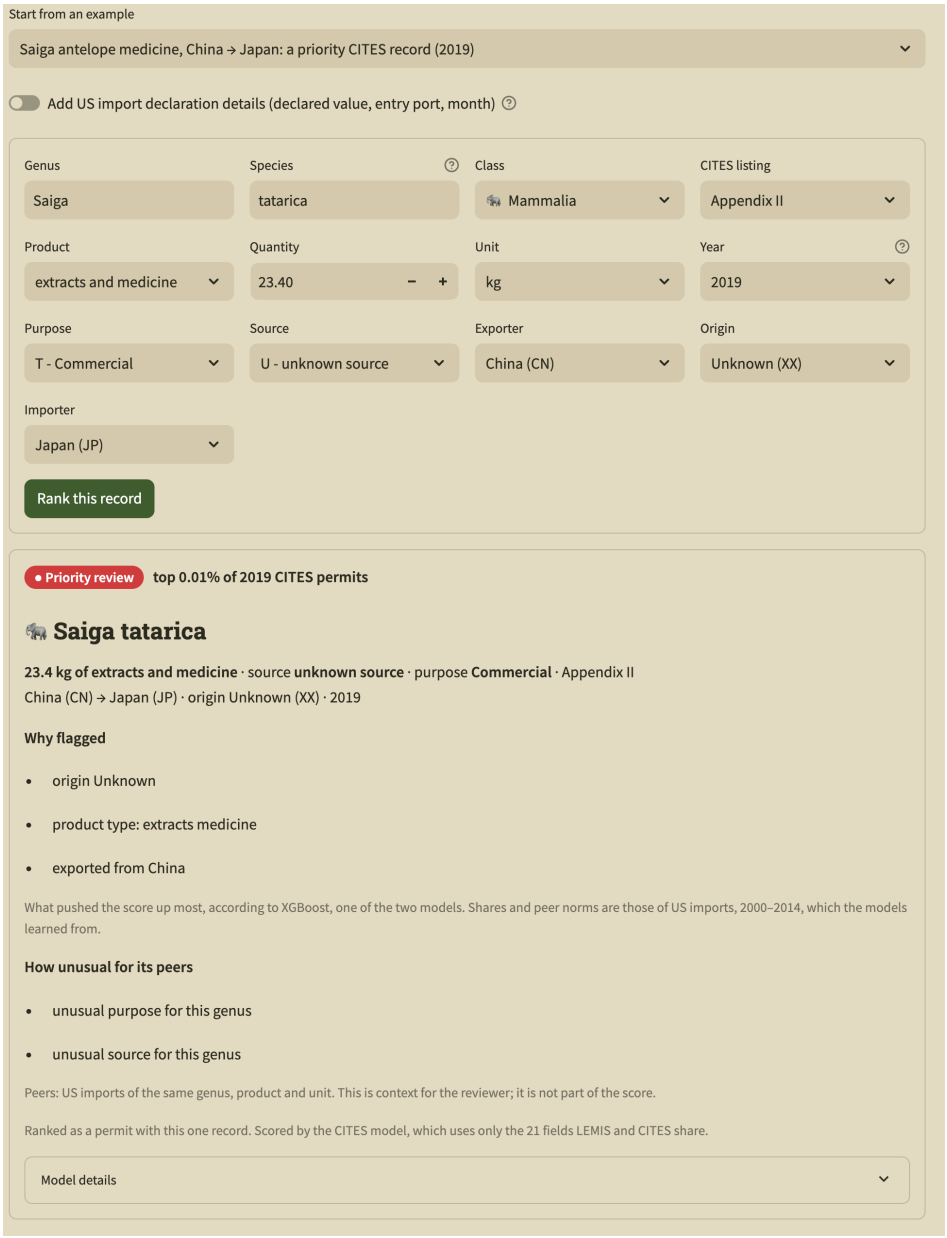


Figure 8 – Check a record: a saiga-antelope medicine record from China to Japan ranks in the top 0.01% of 2019 CITES permits.

5. Tools and techniques

Area	What we used	Module aspect
Data engineering	DuckDB (streaming SQL over 6 GB of CSV), pandas, PyArrow / Parquet	—
Knowledge modelling	LEMIS ↔ CITES crosswalk (product groups, units, source, purpose, countries, higher taxa, the CITES listing in force); rule D for mirror records	Sense making
Feature engineering	Robust peer-group statistics (median/MAD z-scores), rarity and share features; feature selection by groups (leave one group out, mutual information, PSI)	ML techniques
Supervised learning	Logistic regression (L-BFGS and SGD), naive Bayes, decision	Supervised learning;

Area	What we used	Module aspect
	tree, random forest, AdaBoost, XGBoost; time-ordered cross-validation; class-imbalance options (class weights, undersampling, SMOTE-NC, balanced random forest, RUSBoost)	ML techniques
Unsupervised learning	Peer-deviation score, Isolation Forest, K-Means (silhouette, stability), Gaussian mixture (BIC), DBSCAN	Unsupervised learning; ML techniques
Hybrid / ensemble	Bagging and boosting; forward selection of a stack; rank averaging against a logistic meta-learner	Hybrid ML / ensemble
Transfer / domain shift	Domain classifier on raw fields, PSI, PCA projection, regional holdouts, ageing windows	ML techniques
Evaluation	PR-AUC with shipment-bootstrap intervals, ROC-AUC, cases found and precision at inspection budgets per shipment, capture curves, triage bands	—
Sense making	TreeSHAP reasons, peer context, permit-level review queue, US-law warning, three checks against outside evidence	Sense making
Application	Streamlit, Plotly, Matplotlib	—
Tooling	Python 3.11, scikit-learn 1.5, XGBoost 2.1, imbalanced-learn 0.12, SciPy, joblib, Git/GitHub	—

The project develops, integrates and demonstrates the first three required aspects (supervised/unsupervised learning, machine-learning techniques, hybrid/ensemble). The explanation and triage layer adds a sense-making component on top.

6. System performance

6.1 Evaluation protocol and its validity

- **The one test.** All results in §6.2–6.6 come from LEMIS 2013–2014 (783,365 line items, 1.58% enforcement cases), scored once after every design choice (§4.3). The models were trained on 2000–2010, and the triage bands were fixed on 2012.
- **Four held-out sets.** All line items; species with a CITES listing in force in the shipment year (234,574 line items, 2.02% cases), the part of LEMIS most like CITES, on which RQ2 is answered; commercial imports; and listed commercial imports.
- **Two units.** Per line item, and per shipment: a shipment is scored by its highest line and counts as a case if any of its lines is one. Inspection effort is measured in shipments, the unit an inspector opens.
- **Metrics.** PR-AUC with 95% intervals from 1,000 bootstrap draws of whole shipments; ROC-AUC; the share of cases found at budgets of 1%, 5% and 10%; and the share that must be inspected to find 50% or 80% of cases.
- **Baselines.** Random inspection, and a “most endangered first” listing rule (Appendix I, then II and III, wild-caught first within each) using the listing in force in the shipment year.

Validity of the model selection. Done right: the split is by shipment year, with training, choice and test years kept apart (Figure 3); hyperparameters were tuned by time-ordered cross-validation inside the training years; peer norms were learned only on each fold’s training part; the appendix feature uses only listings recorded up to the shipment year; a shipment’s line items are never split; declared value, which may be recorded after the decision, is not used by the deployed model; and the test result did not change the design.

Caveats, stated plainly: (1) the training population and the mirror-record rule were first decided on the test years, in notebooks 04 and 05, before this protocol existed; both were rechecked afterwards (§3.4), but the original

decisions leak mildly into the test. (2) During tuning, XGBoost stopped early on the same fold data it was scored on, which flatters it slightly in the search; the choice itself was made on 2011. (3) Two choices came after seeing a 2011 result: the tie in the supervised comparison was broken on deployment cost, and the forest was shrunk for the hosted app. None came after the test. (4) The training-population check, rerun after the test, favoured listed species on PR-AUC; it is reported, not acted on. Refitting the deployed model on 2000–2014 after the test is not a problem: it changes the training years, not the design.

6.2 Main results (RQ2)

Table 1 – The one test: LEMIS 2013–2014, species with a CITES listing in force. The full ensemble adds LEMIS-only declared value, port and month and is not deployed; the random baseline is one random ordering.

Model	PR-AUC per line item (95% interval)	PR-AUC per shipment	ROC-AUC per shipment	Found at 5% of shipments	Inspect to find 80%
WildGuard: XGBoost + random forest (deployed design)	0.540 (0.519–0.559)	0.643	0.903	63.4%	20.6%
XGBoost alone	0.546 (0.525–0.565)	0.649	0.908	63.5%	16.2%
Random forest alone	0.509 (0.489–0.529)	0.610	0.887	62.6%	26.5%
Logistic regression	0.470 (0.448–0.491)	0.567	0.887	61.7%	21.4%
WildGuard + LEMIS-only fields (full)	0.556	0.659	0.910	64.3%	18.1%
Listing rule (most endangered, wild first)	0.026 (0.025–0.028)	0.039	0.494	5.8%	80.3%
Random inspection (one random order)	0.020	0.035	0.472	4.3%	81.5%

Every learned model beats both baselines by a wide margin. On the records most like CITES, WildGuard reaches a PR-AUC of 0.540 per line item, 27 times random, against 0.026 for the listing rule and 0.020 for random inspection, and 0.643 per shipment. Across all line items it is 0.597 (0.582–0.613).

- **The obvious manual heuristic barely works.** Inspecting the most endangered, wild-caught species first gives a PR-AUC of 0.026, 1.3 times random. Listing status alone is not the risk signal; the combination of fields is. This is the core argument for a pattern-recognition approach.
- **One result went against the design.** XGBoost alone scored slightly higher than the two-model ensemble: −0.006 per line item (interval −0.009 to −0.003), a significant difference, with equal recall at the review budgets (63.4% against 63.5% of case shipments at 5%). On 2011 the ensemble had been ahead by 0.006. The ensemble was chosen before the test and kept, so the test stays a test; we report the result as it came out.
- **Commercial trade is harder.** On commercial imports the deployed model's PR-AUC is 0.258 per line item (0.236–0.278), 31 times random, and 0.214 on listed commercial imports; XGBoost alone is again slightly higher (0.278 and 0.234). Personal imports, half of LEMIS's enforcement cases, are easier (PR-AUC 0.92 at a 25% base rate). Because CITES is 90% commercial, the commercial figures are the better guide to CITES-like trade.

6.3 What an inspection budget buys (RQ3)

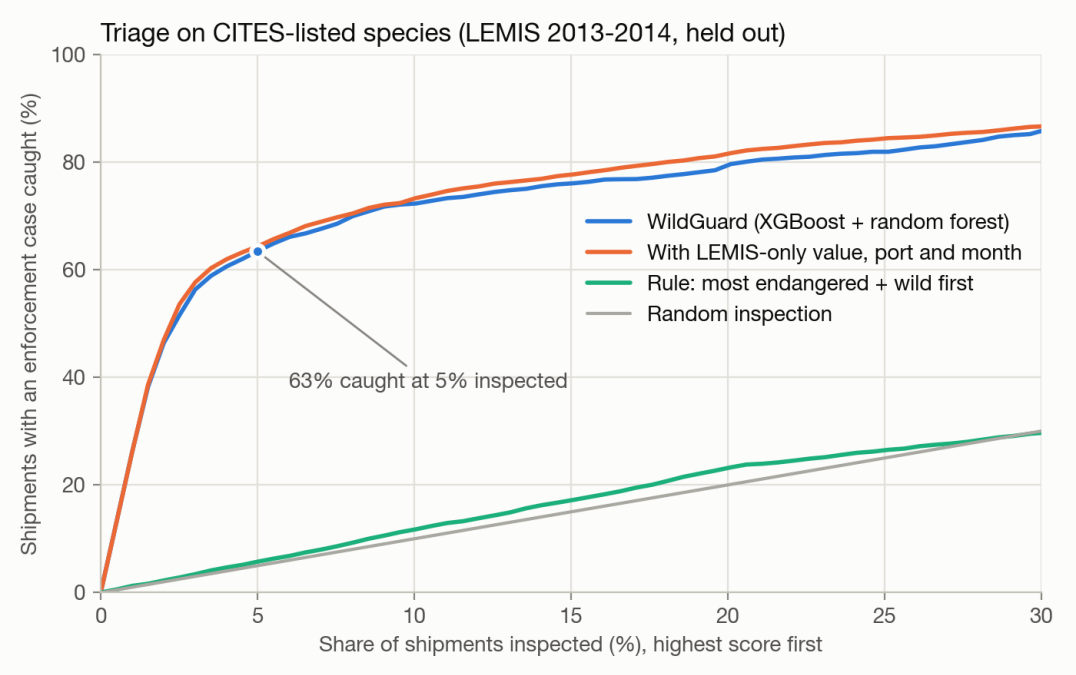


Figure 9 – Share of shipments with an enforcement case found against the share of shipments inspected, highest score first (CITES-listed species, 2013–2014).

Table 2 – What an inspection budget buys, per shipment (CITES-listed species, 2013–2014: 94,147 shipments, 3.8% with an enforcement case). "Found" = share of the shipments with a case that fall within the budget; precision = share of inspected shipments that had a case.

Strategy	Top 1%: found (precision)	Top 5%: found (precision)	Top 10%: found (precision)	Inspect to find 50%	Inspect to find 80%
WildGuard	26.1% (99.4%)	63.4% (48.3%)	72.2% (27.5%)	2.3%	20.6%
Listing rule	1.3% (4.8%)	5.8% (4.4%)	11.7% (4.5%)	52.6%	80.3%
Random inspection	0.7% (2.8%)	4.3% (3.3%)	8.8% (3.4%)	54.2%	81.5%

Framed as an inspector's decision: with a budget of 5% of shipments, WildGuard finds 63% of the shipments with an enforcement case, against 5% for random inspection and 6% for the listing rule. Finding 80% of them takes 21% of shipments instead of 80%, and 99% of the top 1% of shipments had a case. Across all shipments, not only listed species, the same budget finds 73%, and 80% of the cases take 9% of shipments.

Triage bands. Cut into three bands with thresholds fixed on 2012, the deployed score auto-clears 54.9% of 2013–2014 line items while missing 6.0% of enforcement cases (an enforcement rate of 0.17%). The priority band holds 1.9% of line items and 59.0% of cases, almost one in two of them a real case (48.0%). The review band between them (43.1% of line items, 34.9% of cases) is where human judgement is spent (Figure 10).

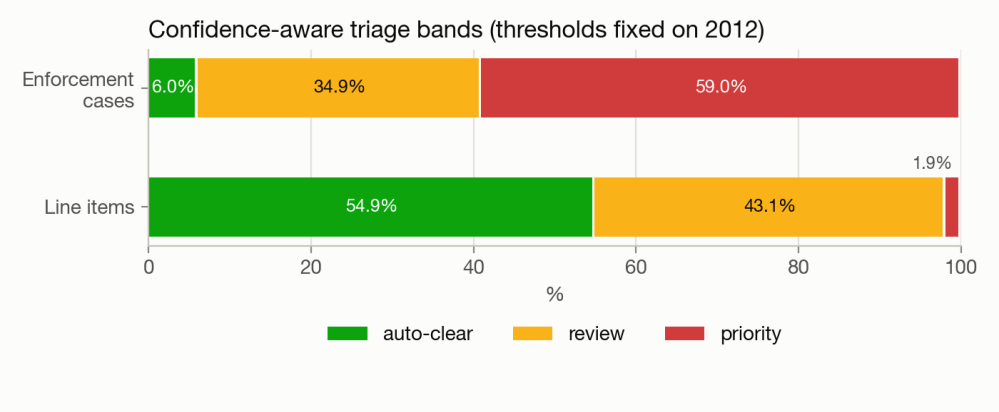


Figure 10 – Triage bands, thresholds fixed on 2012 and applied to 2013–2014.

6.4 Does combining supervised and unsupervised evidence help? (RQ4)

Combining supervised models helped a little on 2011; adding unsupervised evidence did not help at all. Adding the random forest to XGBoost raised PR-AUC on 2011 by 0.006 (interval +0.003 to +0.010), and no unsupervised detector improved the pair (§4.5). The earlier design's stack of XGBoost with peer deviation, Isolation Forest and K-Means was no better than XGBoost alone on the same footing (+0.0004, interval −0.003 to +0.004). On the one test the combination did not hold up: the ensemble scored 0.006 below XGBoost alone (§6.2, Figure 11). The answer to RQ4 is no for unsupervised evidence, and at best a small, unstable gain for combining the two supervised models.

Unusual is not illegal. Each detector alone ranks enforcement cases above random (peer deviation 0.151 against 0.024 for random on listed 2011 line items), so the detectors do find something. But a great deal of legal trade is statistically unusual, and the supervised models already capture what the detectors see. This is why the transferred labels matter, and why the peer-deviation score is context on the flag card rather than part of the score.

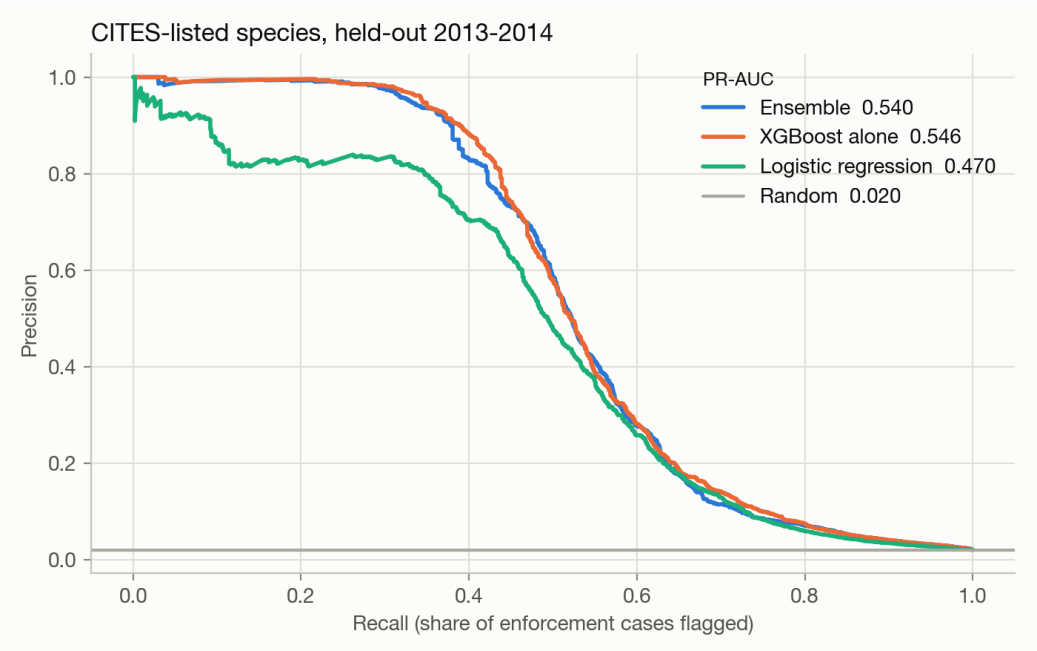


Figure 11 – Precision–recall curves on the one test (CITES-listed species, per line item): the ensemble, XGBoost alone, logistic regression and random ranking.

What the model relies on (Figure 12). Declared purpose carries 24% of XGBoost's split gain, followed by exporter country (15%), origin country (12%), product form (11%), genus (9%) and the rarity of the product, purpose and source combination (6%). The model has learned contextual risk: what is declared, for which species, and from where, relative to the norm.

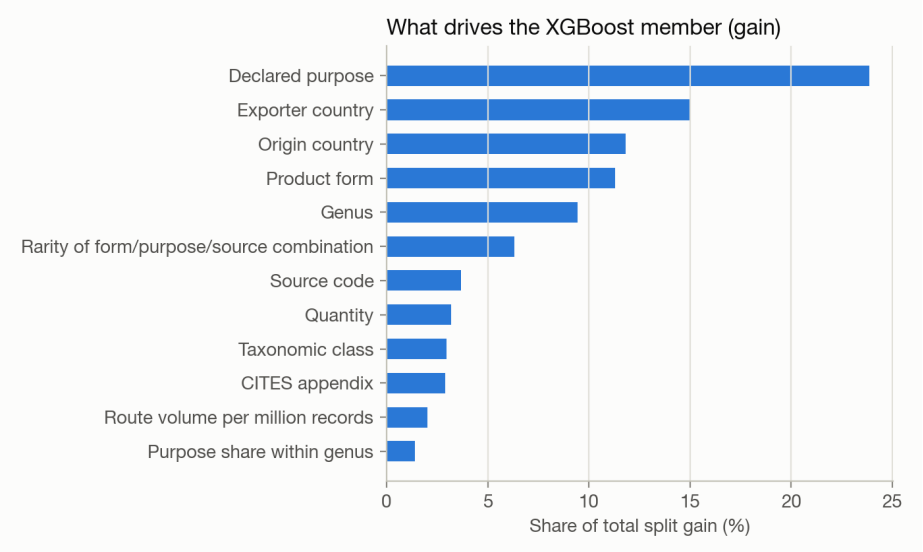


Figure 12 – Feature importance of the deployed XGBoost member (share of total split gain).

6.5 Robustness checks

Table 3 – Robustness on the one test (LEMIS 2013–2014). Lift = PR-AUC per line item ÷ base rate.

Subset	Base rate (line items)	PR-AUC per line item	Lift over random	PR-AUC per shipment	Found at 5% of shipments	Listing rule PR-AUC
All line items	1.58%	0.597	38×	0.708	73.3%	0.039
CITES-listed species	2.02%	0.540	27×	0.643	63.4%	0.026
Commercial	0.82%	0.258	31×	0.307	53.7%	0.016
Listed and commercial	1.04%	0.214	21×	0.278	43.6%	0.014

- **What the LEMIS-only fields add.** Declared value, entry port and month exist only in LEMIS. On the 208,620 test line items of 2013 with a recorded value (2014 has none), the same ensemble with them reaches PR-AUC 0.638 against 0.603 without: +0.035 (interval +0.030 to +0.041). That is what aligning the two datasets gives up (RQ1).
- **Stable over the test years, with drift.** PR-AUC on listed line items is 0.574 for 2013 and 0.512 for 2014, consistent with the ageing measured in §4.3.
- **Calibration.** XGBoost's probabilities are reasonably calibrated on LEMIS: in the highest decile of predicted risk the mean prediction is 18.5% and the observed enforcement rate 14.1%. On CITES the score is used only as a ranking.
- **Worked example.** The Indian star tortoise shipment of §1.2, a training-year record and so an illustration rather than a test, ranks above 99.2% of the 2013–2014 line items. Its reasons are its declared personal purpose (5.9% of *Geochelone* records), its taxon and its rare route from Malaysia. The highest-ranked held-out enforcement cases are equally recognisable: sea-turtle and leopard "medicine" carried as personal imports from Mexico and China, and bear medicine from China, all seized or abandoned.
- **Reproducibility.** The CITES scoring is not bit-for-bit reproducible: the random forest sums its trees in parallel, so ranks move by up to 0.0006 between runs; in one rerun 4 of 46,734 priority records changed tier.

6.6 Transfer robustness (RQ5)

CITES has no labels, so transfer to CITES cannot be scored directly. We break the shift into four parts and measure what can be measured (Table 4).

Table 4 – The parts of the shift from LEMIS to CITES, and how each was measured.

Shift	Measured by	Result
Time	Five-year training windows scored on 2011 (notebook 13)	PR-AUC 0.570 at a 1-year gap, 0.512 at 7 years; the deployed model is refit on 2000–2014, and later CITES years are likely ranked less well
Trade mix	The four held-out sets (§6.2, §6.5); five regional holdouts	Commercial and listed commercial trade much harder; no significant loss against XGBoost alone on any held-out region
Recording	Domain classifier on raw fields: LEMIS listed species against the CITES records the US reported for the same 2013–2014 imports	AUC 0.83: even the same trade, recorded twice, is distinguishable, mainly by appendix (PSI 0.94), class (0.25) and source (0.21)
Jurisdiction	Not measurable: no enforcement outcomes outside the US	Stated as a limitation; seen indirectly in §6.7 (marine mammals, Siamese crocodile)

The deployment shift is large. A domain classifier separates LEMIS training records from the ranked CITES records with AUC 0.985. The biggest differences are in the CITES appendix (PSI 6.3; most LEMIS line items are of species CITES does not list), class (1.9) and product form (1.7); unit, origin region and source shift much less (Figure 13). Almost every feature shifts strongly, but no feature group could be dropped for transfer: each remaining group significantly helps on the listed held-out set (notebook 12).

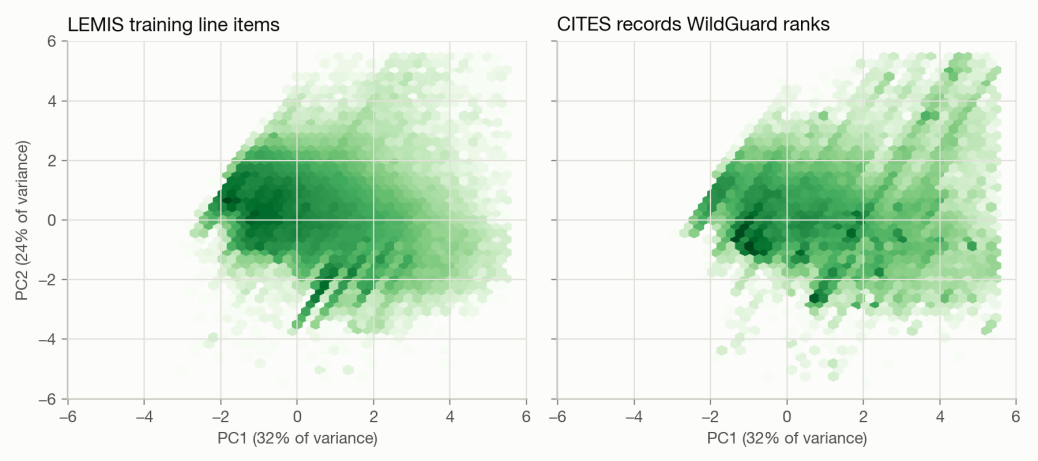


Figure 13 – LEMIS and CITES records projected on the first two principal components of the deviation features.

Adaptation steps tested and rejected. No label-free adaptation step was adopted: peer norms learned from the target were significantly worse on two of six held-out populations (§4.2); importance weighting toward CITES lowered PR-AUC on held-out CITES-listed records from 0.630 to 0.600 in the earlier pipeline; and no shifting feature group could be dropped. A labelled set in CITES's own format, built from the CITES records the US reported for 2013–2014, proved unusable, because most enforcement cases never reach the CITES record (§1.4).

A methodological lesson. Our first domain classifiers used every model feature and separated every pair of populations perfectly (AUC 1.000), even two samples that should look almost alike. The cause was **fingerprinting**: group-level statistics such as a route's share of a genus's trade take one exact value per group within each population, so a tree model recognises the population from those values rather than from any real difference in

trade. That also made the importance weights degenerate. Restricting domain classifiers to raw fields fixed both problems. We report this because it would silently invalidate a shift analysis.

6.7 Applying WildGuard to CITES (RQ6)

Scale. WildGuard ranks the 5,465,862 CITES animal-trade records of 2015–2024 that rule D selects, after removing Appendix N and setting aside 33,053 records already marked as confiscated or for law enforcement. They sit on 3,274,789 export permits, ranked within each year: the priority tier holds 32,741 permits (41,057 records), about 3,300 permits a year worldwide.

Where the priority tier concentrates. Across exporter-reported records, 0.73% sit in priority permits (Table 5).

Table 5 – Share of exporter-reported records in priority permits, by product and by source (CITES 2015–2024).

Product	In priority permits	Source	In priority permits
bone	12.1%	D – bred in captivity (Appendix I, commercial)	29.13%
shells	12.0%	U – unknown	22.02%
meat	8.5%	not declared	3.93%
fins	6.1%	W – wild	0.36%
teeth and claws	5.9%	O – pre-Convention	0.31%
body parts	5.5%	C – bred in captivity	0.04%

Worked and edible animal parts and unknown sources are flagged most, and personal-purpose records far more than commercial ones (5.1% against 0.6%). Appendix I taxa are flagged at 11.0%, Appendix II taxa at 0.2%. Pre-Convention claims, which the earlier pipeline flagged heavily before the crosswalk mapped LEMIS's source P to CITES's O, now sit at 0.3%: the model has learned what normal pre-Convention trade looks like.

Half of the priority tier is one pattern. Siamese crocodile (*Crocodylus siamensis*) makes up 48.5% of priority records but 0.6% of ranked records: mostly leather from Thailand to China and Japan, bred in captivity under Appendix I (source D). In LEMIS, 95% of the 284 Siamese crocodile imports were refused, consistent with the species' listing as endangered under the US Endangered Species Act. The model has learned a US import prohibition and applies it to trade that CITES permits between other countries. Marine-mammal products show the same effect on a smaller scale: traded between countries other than the US, they are 8.6% of priority records but 0.4% of ranked records, and the Greenland → Denmark route is flagged in 67% of its records. Their flag cards carry a US-law warning (§4.6); Siamese crocodile does not, because the warning was designed before this pattern was measured (§9).

Three checks were fixed before the rebuilt pipeline ran (`DESIGN_QUESTIONS.md`, C7). A pass means the ranking is consistent with outside evidence, not that it is accurate.

Check 1 – consistency with UNODC seizure statistics: fail. Animal groups over-represented in global seizures relative to their legal trade should be flagged more often (Table 6, Figure 14). Across the 14 animal groups of the *World Wildlife Crime Report 2024*, the Spearman correlation between the seizure-to-trade ratio and WildGuard's priority share is $\rho = 0.13$ (one-sided permutation $p = 0.32$), below the listing rule's 0.43; the pass mark was $p < 0.05$ and above the listing rule. Pangolins, by far the most over-seized group relative to legal trade, have only 133 ranked records and none in a priority permit. The check first passed ($\rho = 0.54$, $p = 0.025$), but that result rested on three pangolin records that reached the priority tier only because their permit IDs collided with unrelated trades. After the permit key was corrected (§4.7), before the check was rerun, it fails. Without pangolins, which the check already flagged as unstable, ρ is 0.42; that figure is computed after seeing the result and is not the criterion.

*Table 6 – Consistency with UNODC seizure statistics (exporter-reported CITES records 2015–2021). UNODC seizure shares are the animal groups of WWCR 2024 Fig. 2.1, renormalised to sum to 100%; the seizure-to-trade ratio is the seizure share divided by the group's share of legal CITES records. * fewer than 500 records, flagged as unstable before the check.*

Animal group (UNODC)	CITES records 2015–2021	UNODC seizure share	Seizure-to-trade ratio	WildGuard priority share	Listing-rule top-1% share
Pangolins *	133	3.1%	774.67	0.00%	54.13%
Bivalve molluscs	4,232	9.2%	73.04	0.71%	0.07%
Elephants	21,918	9.2%	14.10	0.53%	15.83%
Sea snails	9,063	3.1%	11.37	9.07%	0.02%
Turtles and tortoises	25,085	6.2%	8.21	0.25%	3.81%
Carnivores	80,913	7.7%	3.18	1.83%	6.35%
Parrots and cockatoos	88,155	6.2%	2.34	0.18%	3.01%
Even-toed ungulates	45,942	3.1%	2.24	1.25%	0.97%
Corals	600,817	24.6%	1.37	0.07%	0.00%
Sturgeons and paddlefishes	139,326	3.1%	0.74	0.02%	0.01%
Birds of prey	79,534	1.5%	0.65	0.35%	16.40%
Lizards	189,172	3.1%	0.54	0.10%	0.07%
Snakes	415,204	6.2%	0.50	0.17%	0.01%
Crocodilians	1,649,020	13.8%	0.28	1.07%	0.46%

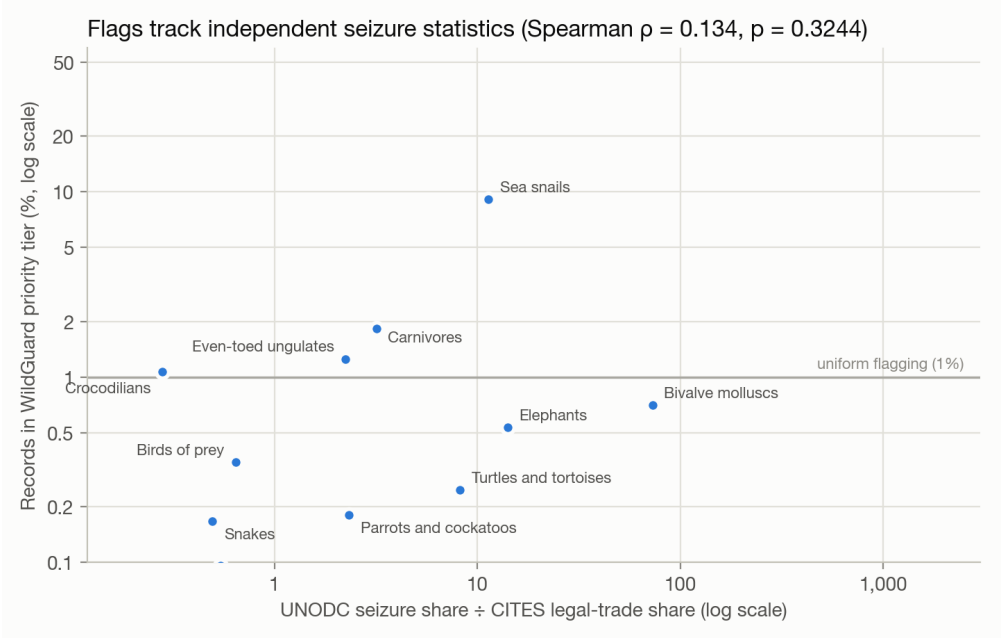


Figure 14 – WildGuard priority share against the seizure-to-trade ratio of each UNODC animal group.

Check 2 – the confiscated-specimen proxy: pass. CITES marks 32,750 records in this population as source I (confiscated or seized specimens). Retrained without any source-derived feature, the model ranks them far above the rest (ROC-AUC 0.888, 95% interval 0.886–0.891; the pass mark was a lower end above 0.5), with 66% of them in its top 5%. Patterns learned from US enforcement outcomes point toward material that CITES independently records as enforcement-related.

Check 3 – the top 50 taxa: fail. The highest-ranked priority record of each of the top 50 taxa was put into one of three categories fixed in advance, by a stated rule (`validate_cites.classify`) rather than by two human reviewers: (a) a recognised risk pattern (worked or high-value parts, personal-purpose specimens of Appendix I

taxa, unknown or pre-Convention source, unknown origin): 62%; (b) a US legal regime carried abroad: 8%; (c) no apparent reason: 30%. The pass mark was at least 50% in (a) and at most 20% in (c). All 15 records in (c) are personal-purpose imports of wild Appendix II specimens, 12 into the US and 3 into New Zealand: a pattern US enforcement acts on (25% of US personal-purpose test line items were enforcement cases), but not one of the patterns fixed in advance (Table 7).

Table 7 – The ten highest-ranked records of check 3 (one per taxon), with their category: (a) recognised pattern, (b) US law abroad, (c) no apparent reason. Purpose: T commercial, P personal. Source: W wild, U unknown.

Year	Taxon	Product	Quantity	Purpose / source	Route	Category
2021	*Iguana spp.*	bodies	30	P / W	MX → US	(c)
2017	*Iguana iguana*	meat	1.18 kg	P / W	MX → US	(c)
2016	*Strombus gigas*	meat	0.454 kg	P / W	HT → US	(c)
2020	*Scleractinia spp.*	raw corals	1	P / W	PH → US	(c)
2019	*Saiga tatarica*	medicine	23.4 kg	T / U	CN → JP	(a)
2018	*Panthera tigris*	medicine	15	P / W	PE → US	(a)
2023	*Eretmochelys imbricata*	carvings	1	P / W	PE → US	(a)
2020	*Hippocampus spp.*	bodies	4	P / unknown	CN → US	(a)
2016	*Alligator mississippiensis*	leather products (small)	1	T / U	CA → BE	(a)
2024	*Moschus spp.*	medicine	40	T / W	VN → US	(a)

The ranking is not simply US-like trade. US imports make up 22.3% of ranked records but 6.6% of priority records, while personal-purpose records are 27.5% of priority records against 3.7% of ranked ones. The personal-import pattern is what US enforcement most often acted on, and it carries over.

7. Findings and discussion

1. Enforcement outcomes contain learnable structure that partly transfers. Using only fields CITES also records, the models find enforcement cases on held-out LEMIS with a PR-AUC of 0.540 on CITES-listed species, 27 times random, and find 63% of the shipments with a case at a 5% budget. What they rely on is context: declared purpose, exporter, origin, product and taxon, and how rare that combination is. Single fields are weak, and the "most endangered first" rule is barely better than random. The case for pattern recognition here is that combinations of fields carry the signal.

2. The practical value is triage. For an inspector with a fixed budget, WildGuard turns 5% of inspections into 63% of the case shipments found, and finding 80% takes 21% of shipments instead of 80%. In band form, more than half of all line items can be auto-cleared while missing 6% of cases. This meets the objective: a ranked, evidence-grounded shortlist rather than a verdict.

3. Comparison decided the design, and some results went against it. Six supervised candidates were compared under one protocol; boosting and bagging tied, and none of five unsupervised detectors improved them. On the one test, XGBoost alone edged out the chosen ensemble; we kept the design and report it. Unusual is not illegal: the detectors find something, but not what the labels find.

4. Transfer can be measured only in parts. Time, trade mix and recording shifts were measured and are large; jurisdiction cannot be measured without labels outside the US. Label-free adaptation did not help, so a small labelled sample from a CITES authority remains the most valuable next step.

5. On CITES the ranking shows its origin. It finds confiscated specimens it was never told about (check 2), but it does not track UNODC seizure statistics (check 1), and 30% of its top taxa show no recognised pattern (check 3). Half of its priority tier is Siamese crocodile leather, lawful CITES trade that US law prohibits, and marine-mammal products follow the same logic. The honest reading is that WildGuard ranks CITES records by how US enforcement would have treated them. That is useful to a reviewer who knows it and misleading to one who does not, which is why every flag carries reasons and warnings.

6. Most of the risk was in the data engineering. The model is standard; the work that decided whether the results could be trusted was alignment, validation and checking. Changes made during the project changed our conclusions:

- a crosswalk error (LEMIS pre-Convention P never mapped to CITES O) had made pre-Convention claims look risky on CITES;
- the appendix feature's genus fallback gave named species with no CITES record the strictest appendix in their genus;
- domain classifiers that "fingerprinted" populations had to be rebuilt on raw fields;
- an anonymised permit ID shared by unrelated trades had merged them in the review queue, and fixing it turned a passing check into a failing one;
- records already marked as confiscated were filling a fifth of an early review queue and had to be set aside;
- each statistic cited from the literature was checked against its source document.

Why this matters. The CITES Secretariat and WCO guidance asks agencies to target inspections by risk. Today that happens officer by officer. WildGuard shows that the enforcement experience already recorded in one national dataset can be turned into a statistically grounded, explainable ranking at the scale of the global trade database. It also shows which parts of that experience transfer and which reflect the jurisdiction they came from.

8. Limitations and responsible use

WildGuard is designed to support human review, and its limitations follow from that.

1. **Enforcement outcomes are not legality.** The label is what US inspectors did. It includes paperwork refusals, and it misses illegal shipments that cleared undetected. The model therefore learns *what inspectors acted on*, including their priorities and blind spots, not ground-truth illegality. Most enforcement cases never reach CITES at all (§1.4), so a flag means that a record resembles trade that was stopped in the US.
2. **Old, US-centric labels.** LEMIS covers 2000–2014 and US enforcement only, while CITES scoring covers 2015–2024 worldwide. Trade routes, species listings (for example the 2016 transfer of all pangolins to Appendix I) and enforcement practice have all changed since. Performance falls with the gap between training and scored years (§4.3), so the ranking is likely weaker for the later CITES years. CITES is also updated every year; WildGuard follows a new release by rebuilding the pipeline, which re-scores every year and updates the app's data preview with it.
3. **Jurisdiction-specific labels.** Some LEMIS outcomes reflect US law rather than CITES rules. Siamese crocodile imports were refused 95% of the time, consistent with the US Endangered Species Act, and its leather traded between other countries is half of the priority tier; marine-mammal products, refused at ten times the average rate under the Marine Mammal Protection Act, are 8.6% of priority records in trade that does not involve the US (§6.7). Only the marine-mammal case carries a warning. A deployment should add importer-specific rules or warnings for every taxon whose LEMIS outcomes reflect a US prohibition.
4. **Appendix lookup through reported trade.** LEMIS has no appendix field, so each LEMIS species gets the listing in force in its shipment year: the most recent appendix CITES recorded for it in or before that year, with Appendix N ignored, and its genus's listing only when no species is named. The CITES database only shows a species' listing through trade that was reported, so a species with no reported trade in or before

the shipment year gets no listing even if it was listed then. The rebuilt pipeline marks such a named species "genus listed" when its genus has a listed species, and "not listed" otherwise. Synonyms are missed, because the lookup matches names.

5. **Personal vs commercial mismatch.** Half of LEMIS enforcement cases are personal imports, while 90% of CITES records are commercial. On commercial imports alone the deployed model's PR-AUC is 0.258 per line item (31× random) against 0.597 overall, so the headline figures overstate performance on CITES-like commercial trade (§6.2).
6. **No accuracy claim on CITES.** Without labels, the CITES-side checks (§6.7) establish consistency with outside evidence at most, and two of the three fail. Scores on CITES are rankings within an inspection budget, not probabilities. Many trades are reported by both the exporting and the importing country; rule D (§3.4) scores an importer's report only when the exporter has no exact counterpart, but the two reports often differ in some field, so some second reports remain (about 2–3% of the top 1% in the design check).
7. **Permit-based signals cannot be learned from labels.** LEMIS declarations carry no permit number, whereas CITES records carry the export permit that links the two countries' reports of one consignment (§3.3). The model therefore cannot learn from US enforcement outcomes whether two reports that disagree, a trade reported by only one side, or a chain of re-exports signals risk. Such patterns can be shown to an analyst but not validated. Whether a trade is reported by both sides also depends mostly on the reporting countries (§3.3), so a missing report from the other side is not in itself evidence of risk. Permit IDs are also imperfect: they collide across unrelated trades, so a permit is identified by its ID, exporter, importer and year, and a permit number one country reuses on one route in one year can still merge consignments.
8. **Value-field caveat.** Zero declared values are more common among refused LEMIS items (24% vs 10%), possibly because values are recorded after the decision. The *full* model's results may therefore be slightly optimistic. The deployed *transfer* model does not use value.
9. **Bias and feedback loops.** Country and route features encode where enforcement has looked before. Flags concentrate on particular exporters, and acting on them without review could reinforce existing enforcement patterns. Flags must remain prompts for human review; flag rates by country should be monitored; and no automatic action should be taken on a score.
10. **A design check after the test.** Rerun on the rebuilt data after the one test, the training-population check favoured training on CITES-listed species only when judged on PR-AUC (+0.007). The deployed system keeps animals with a class, chosen by the rule as written, and the difference is small; it is the first candidate for a retrained version.

9. Future work

- **A small labelled sample from a CITES authority.** In the simulated shift, 5% of target labels recovered more than half of the gap to a fully supervised target model (§6.6). A few thousand reviewed CITES records would be the most valuable next step.
- **Newer LEMIS data.** Freedom of Information Act releases after 2014 would shrink the temporal gap to the CITES scoring window.
- **Mirror-record discrepancies.** In the rerun, the two countries' reports of a permit disagreed only slightly more often among high-risk pairs (11.9% against 9.9%), so disagreement is shown as context, not used as a signal; testing it as a risk signal would need labels.
- **Analyst feedback loop.** Each reviewed flag becomes a new label, which allows active learning and periodic retraining. This turns the triage tool into a learning system.
- **Fairness monitoring.** Track flag and confirmation rates by exporter country and taxon, and calibrate per group where reviewed labels allow.
- **Taxonomic hierarchy features.** Learned embeddings of family and order would let rare genera borrow strength from their relatives.

- **Warnings for every US prohibition.** Extend the US-law warning beyond marine mammals to taxa such as Siamese crocodile whose LEMIS outcomes reflect US import prohibitions, or rank such trade separately when the importer is not the US.
- **Training on listed species, re-tested.** Retrain on CITES-listed species only and test it on a held-out period that has not been used (§3.4).
- **Each CITES release.** Rebuild with every annual CITES release, and track how the priority tier shifts between releases.

10. Conclusion

WildGuard set out to answer whether real enforcement outcomes recorded in LEMIS can be used to prioritise review in the much larger, unlabelled CITES database. On the six research questions:

- **RQ1 – can the datasets be aligned?** Yes, for 21 shared fields, with a crosswalk that corrects codes that look the same but are not. What cannot be aligned costs 0.035 PR-AUC (declared value, port and month), and permits exist only in CITES.
- **RQ2 – does it work?** Yes. On the one test, PR-AUC is 0.540 per line item on CITES-listed species (random 0.020), 0.643 per shipment, and 0.597 on all line items.
- **RQ3 – is it useful?** A 5% budget of shipments finds 63% of the shipments with an enforcement case; finding 80% takes 21% of shipments instead of 80%.
- **RQ4 – does combining help?** Not with unsupervised evidence: no detector improved the ranking. Combining XGBoost with a random forest helped on 2011 but not on the test.
- **RQ5 – does it transfer?** Partly, and measurably: the ranking survives held-out regions, but it ages, and the recording and deployment shifts are large. Jurisdiction cannot be measured without labels.
- **RQ6 – what does it find in CITES?** Confiscated specimens it was never told about (check 2 passes), but not the UNODC seizure pattern (check 1 fails), and a priority tier dominated by trade that US law prohibits: a ranking of how US enforcement would treat global trade, with its biases visible.

The project develops and integrates supervised and unsupervised learning, compared under one protocol, boosted and bagged ensembles, and a knowledge model that aligns two datasets, delivered as a runnable review tool with explanations on every flag. Its honest claim is the one stated at the outset: a WildGuard flag is a well-founded prompt for limited human attention, not a verdict.

AI Tool Use Declaration

We used Claude Code (Anthropic) to write and debug the pipeline and application code, run and analyse the design experiments, draft and refine the data guides, design notes and report text, and create the diagrams and slides. We used ChatGPT (OpenAI) to discuss the validity of transferring labels between the two datasets, which raised the domain-adversarial network we considered and set aside (§4.3) and the point that the evidence for transfer to CITES is indirect (§6.6). We are responsible for the content and quality of the submitted work.

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Appendix A — Repository and how to run

A.1 Using the hosted app

The WildGuard review tool is available at https://wildguard-group37.streamlit.app.

1. Open https://wildguard-group37.streamlit.app in a web browser. No account, password or installation is needed. The hosted app runs the same models and data as the repository.
2. If Streamlit says the app has gone to sleep, click the button to wake it. Streamlit Community Cloud pauses apps that have had no visitors for a while; waking it takes about a minute, and the first page then takes a few more seconds while the models and data load.
3. Choose a page from the sidebar on the left: **Overview**, **Review queue**, **Check a record**, **Does it work?** or **What it flags** (§4.8). On a narrow screen the sidebar is collapsed; open it with the arrow at the top left.
4. To open a page directly, add `?page=` and one of `overview`, `queue`, `check`, `evaluation` or `patterns` to the address, for example https://wildguard-group37.streamlit.app/?page=queue.
5. On **Check a record**, pick a worked example under "Start from an example" or enter a record's fields. Adding `&example=` with `tortoise`, `saiga` or `pythons` loads an example directly: https://wildguard-group37.streamlit.app/?page=check&example=saiga opens the saiga-antelope record of Figure 8.

A.2 Running from the repository

```
PRS-PM-Group-37-WildGuard/
├── SYSTEM CODE/          runnable system: wildguard/ package, app/, models/, ou
tputs/
├── DATA/                 data README + citations, download script, codebook, da
ta guides
├── Project Report/       this report (.docx) and its figures
└── Presentation/         first presentation (PDF), final presentation(PDF)
```

```
pip install -r "SYSTEM CODE/requirements.txt"
streamlit run "SYSTEM CODE/app/app.py"          # MVP, runs from committed artef
acts
python DATA/download_data.py                    # raw data (LEMIS automatic, CIT
ES manual)
bash "SYSTEM CODE/run_pipeline.sh"              # full rebuild, ~40 minutes
```